

may not be required if an acid such as an acrylic acid is used as a monomer in the system.

Dynamic light scattering (DLS) of latex particles

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PNIPAM latex particles samples were prepared with varying concentrations of crosslinker (i.e. N,N'-methylenebis(acrylamide) (BIS)) and surfactant (i.e. sodium dodecyl sulfate (SDS)). A summary of the composition present (wt% based on the monomer latex particles) in the samples and their corresponding aggregation percentage (Aggr %) is provided in Table 1 below. The z-average particle size (DLS Z-Ave) and polydispersity index (PDI) results of the latex particles obtained from DLS experiments are also provided in Table 1. To test for reversibility in temperature response, the measurements were first taken at (i) a temperature of about 40°C, then at (ii) a temperature of about 20°C and finally (iii) at a temperature of about 40°C. As shown in Table 1, the average particle size recorded at temperature (iii) for the samples remains relatively similar to those recorded at temperature (i). Therefore, the temperature response was proven to be reversible.

20 Table 1. Summary of composition and DLS results of PNIPAM samples synthesized for hetero-coagulation studies

Temperature				40°C		20°C		40°C	
%(wrt to monomer)	BIS (%)	SDS (%)	Aggr (%)	DLS Z-ave (nm)	PDI	DLS Z-ave (nm)	PDI	DLS Z-ave (nm)	PDI
Sample A (PNIPAM 3)	1	0.267	0	203.7	0.038	453.6	0.086	204.7	0.010
Sample B (PNIPAM 4)	0.2	0.267	0	207.2	0.014	494.6	0.144	209.6	0.031
Sample C (PNIPAM 5)	0	0.267	0	210.9	0.025	513.3	0.151	212.4	0.024
Sample D (PNIPAM 6)	1	-	48.1	553.5	0.104	20230.0	1.000	806.5	1.000